

Education

Bachelor of Technology (B.Tech) – Electronics & Communication Engineering

CGPA: 6.58

Nov 2022 – Present

Senior Secondary (Class XII): 90%
2022

Experience

Machine Learning Intern

June 2025 – July 2025

- Collaborated with an autonomy team to enhance **GPS-denied navigation systems**, contributing to real-time decision-making logic for indigenous drones.
 - Optimized data pipelines using **CVAT and Roboflow**, ensuring high-quality annotation consistency that improved downstream model accuracy for aerial object detection.
 - Facilitated **edge deployment workflows**, bridging the gap between large-scale training datasets and embedded inference requirements.
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Skills

Languages

Python, C/C++, JavaScript, SQL

Frameworks & Libraries

Flask, HTML, CSS, XGBoost, LightGBM, NumPy, Pandas

Concepts & Architecture

REST APIs, OAuth 2.0, Data Structures, Object-Oriented Programming, Computer Vision

Developer Tools

Git, Docker, Figma, MATLAB, LTSpice, CVAT, Roboflow

Soft Skills

Critical Thinking, Problem Solving, Time Management

Projects

Full-Stack Music Player Application

Full-Stack Personal Project (2025)

- Built a scalable full-stack application using **Flask and Vanilla JavaScript**, implementing secure OAuth 2.0 flow for third-party music API integration.
- Engineered **real-time playback synchronization and queue management**, ensuring state consistency across browser sessions.

- Designed a responsive, vinyl-inspired UI with asynchronous DOM manipulation to deliver smooth interaction for music discovery and album exploration.
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Hybrid Framework for Motor Imagery EEG Classification

Research Paper (Accepted – 2026)

- Developed a hybrid **Brain-Computer Interface (BCI)** framework combining handcrafted signal features (FBCSP, tangent-space) with deep learning embeddings (EEGNet).
 - Engineered an **ensemble meta-classifier** (XGBoost, LightGBM, Logistic Regression) fine-tuned on subject-specific data, achieving **93.16% accuracy** on the BCI Competition IV-2a dataset.
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Relevant Coursework

Analog Electronics

Digital Electronics

Digital Signal Processing

VLSI Design

Embedded Systems

Computer Architecture

Certifications

- Computer Science Foundations Certification
 - Python Programming Certification
 - Cybersecurity Fundamentals Certification
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Leadership & Volunteering

Volunteer Educator

Feb 2024 – April 2024

- Conducted sessions on **computer basics and academic subjects** for more than 50 underprivileged children.
- Designed simplified learning modules to improve **digital literacy and engagement**.

Event Volunteer – Physics Outreach Event

Nov 2023

- Managed on-ground logistics and participant coordination for a university-level science outreach event.
- Collaborated with the organizing committee to execute engagement activities and improve participation.